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Final Project Report:

Global Housing Market Dynamics Amid the COVID-19 Pandemic: A Data Driven Analysis

For this data science project, I explored how the COVID-19 pandemic may have influenced global housing markets by comparing key indicators from the years 2020 and 2023. Using a real-world dataset titled *"Global Housing Market Analysis (2015–2024)"* sourced from Kaggle, I analyzed multiple factors such as the House Price Index, Affordability Ratio, Rent Index, GDP growth, inflation, population growth, and urbanization across 20 countries. For clarification on the indicators: the House Price Index represents the average price of houses in USD, while the Rent Index indicates the median monthly rent for properties in USD. The Affordability Ratio compares the median house price to the median household income. GDP growth, inflation rate, urbanization rate, and population growth are all expressed as annual percentage rates.

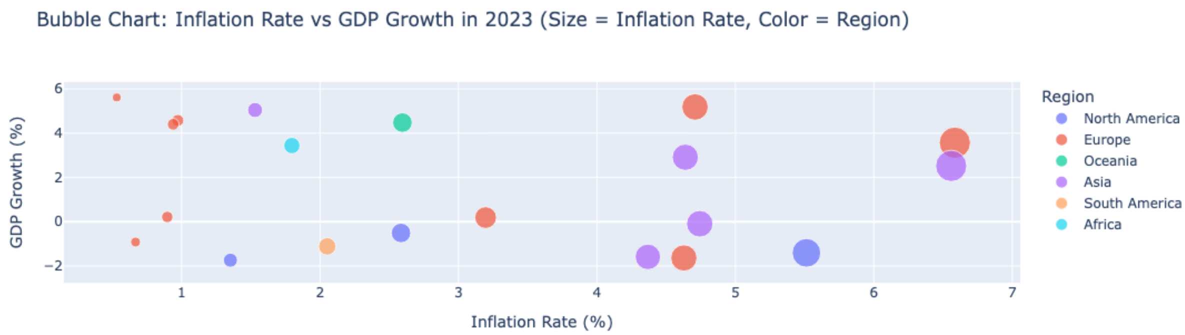
As a college student interested in how external shocks, such as a global pandemic, can affect economic systems, I wanted to examine whether trends in housing affordability and market behavior shifted during this time. To keep the analysis focused, I narrowed the dataset to just two years: 2020, the onset of the pandemic, and 2023, a point of global economic recovery. While the dataset offers helpful economic insight, it includes a relatively small and unevenly distributed sample of countries, with certain regions more heavily represented than others. For example, there are 9 European countries whereas only one African country (South Africa). Still, this project highlights important housing and economic patterns that may reflect the pandemic's broader impact on global real estate.

I started by exploring the structure of the dataset and found that there were no missing (null) or duplicate values. Even so, I wanted to make sure the data was properly cleaned and ready for analysis. I removed the "Construction Index" column, as it wasn't relevant to the focus of my project. To make filtering by year easier later on, I created a new categorical column called "Year_cat" by copying and converting the original "Year" column. I also verified that all numeric columns made sense—for example, I checked that mortgage rates didn't include any negative values.

Next, I performed a descriptive analysis. After reviewing the summary statistics using the "describe" function, I examined the value counts for the categorical columns "Country" and "Year_cat." I used the "groupby" function to better understand patterns by calculating the mean of metrics like the House Price Index across different countries and years. To support geographic visualizations later on, I created a new column for each country's Alpha-3 code and another column that grouped countries by region (like "North America") for easier comparison. With regions defined, I grouped the data again to look at descriptive statistics across those regions and checked for any unusual patterns or anomalies. Once all of that was complete, I saved the cleaned and enriched dataset to use as the base for further analysis and visualization.

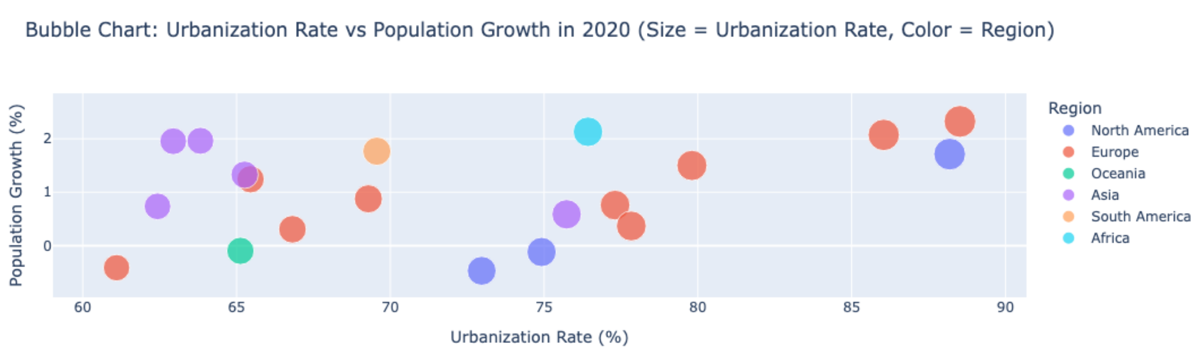
For the main part of my analysis, I decided to focus specifically on the years 2020 and 2023. I chose these two years because they represent clear before-and-after snapshots of the pandemic's structural impact on housing markets. The intermediate years like 2021 and 2022, were more transitional, marked by lockdowns, reopenings, and fluctuating conditions, which could complicate the analysis. By focusing on 2020 as a starting point and 2023 as a more stabilized endpoint, I was able to better isolate long-term changes. I created interactive bubble charts using Plotly for all visualizations, comparing key variables separately for both 2020 and 2023. In each graph, data points were colored by region to help highlight geographical differences, and the bubble size represented the value of the x-axis to give a visual sense of magnitude.

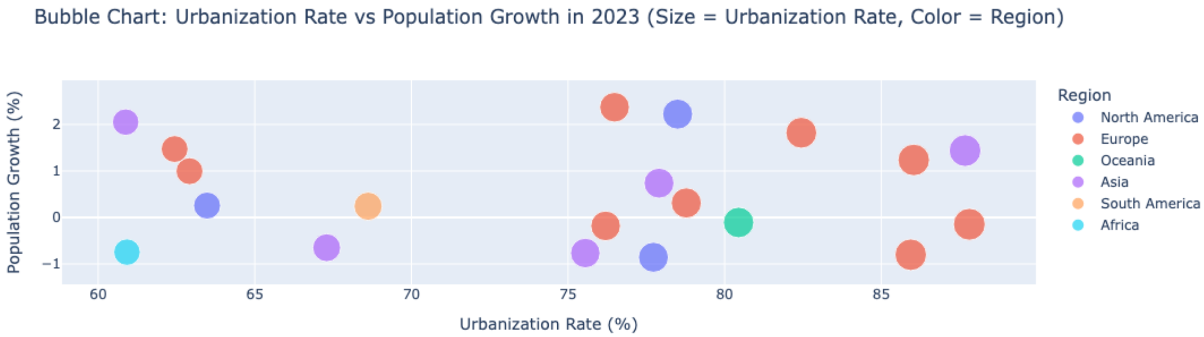
My first bubble chart visualizes the relationship between inflation rate and GDP growth, which is important because it helps highlight how countries are balancing economic recovery with rising prices. By examining both indicators together, we can better understand whether inflation is being driven by healthy economic activity or if it's occurring alongside stagnation or decline, which can signal deeper issues.



The chart reveals an interesting shift between 2020 and 2023. In 2020, many countries experienced negative GDP growth most likely due to widespread lockdowns and disruptions caused by COVID-19. Inflation was also relatively low in many areas perhaps because demand dropped sharply. However, by 2023, although some European countries had recovered and returned to positive GDP growth, a few North American countries appeared to move in the opposite direction—experiencing economic slowdowns despite rising inflation. This suggests that while some economies managed to bounce back, others faced new challenges like high consumer prices and shrinking output, a combination often referred to as "stagflation." Overall, this visualization shows how uneven the global recovery has been. The mix of high inflation and negative GDP growth in 2023 for some countries could reflect ongoing supply chain issues, increased cost of living, and delayed responses to earlier economic shocks.

My next bubble chart visualizes the relationship between urbanization rate and population growth, which is especially relevant when considering the pandemic's influence on global housing trends. The COVID-19 crisis reshaped where and how people live, affecting migration patterns, urban development, and housing demand. By examining how urbanization and population growth evolved from 2020 to 2023, we can better understand how these shifts influenced the pressure on housing markets—particularly in cities where demand may have surged despite slower or even negative population growth overall.

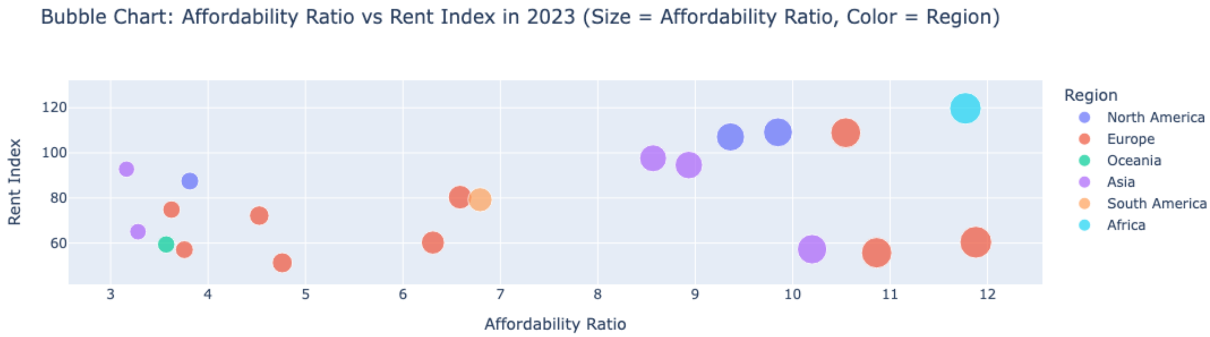




From the chart, we can see that between 2020 and 2023, population growth remained relatively stable on a global scale, but countries such as Brazil in the region of South America, experienced negative growth. Despite this, there was a clear trend toward higher urbanization rates across nearly all regions. This suggests that even in places where population growth slowed or declined, people continued to move into urban areas, reinforcing the idea that cities are still viewed as hubs for opportunity and access.

Looking more closely at this trend, the rise in urbanization may be driven by a combination of factors, including changes in lifestyle accelerated by the pandemic. The shift toward remote work gave many people more flexibility in choosing where to live, often leading them to relocate to smaller cities or suburban areas with more space and lower costs yet still classified as urban environments. Overall, the chart highlights a growing global preference for urban living, even as total population growth plateaus in many regions.

My final bubble chart illustrates the relationship between the affordability ratio and the rent index which are two key measures that reveal how financially accessible housing is in different countries. This comparison is especially important in the context of the COVID-19 pandemic, which disrupted housing markets worldwide and reshaped both rental demand and affordability.



The visualization shows how, in 2020, lower affordability ratios (indicating more affordable housing relative to income) were mostly limited to European countries. By 2023, however, more regions, including parts of Asia and North America, began showing similarly lower ratios, suggesting a shift toward improved affordability in some areas. The graph also shows a general positive correlation between affordability ratio and rent index, meaning that as rent prices increase, so does the strain on affordability. This trend reveals how rising rent costs have increasingly outpaced income growth in many regions, especially as economies recover post-pandemic.

In conclusion, this analysis illuminates the wide-reaching impact of the COVID-19 pandemic on global housing markets, reflected in shifting trends in affordability, GDP growth, inflation, and urbanization. The data reveals how countries responded differently to pandemic-induced disruptions, with some experiencing greater economic strain and housing instability than others. From rising inflation and negative GDP growth to changes in population movement and affordability, the pandemic triggered complex and uneven effects across regions. Although the visualizations offer meaningful insights into these patterns, the analysis is limited by the dataset's scope, which includes a relatively small and regionally unbalanced sample of countries. Expanding the dataset in future studies would provide a more comprehensive view of how the pandemic reshaped housing dynamics around the world and help us understand holistically, the long-term effects of the pandemic not only on the housing market, but our world as we know it.

References

Global Housing Market Analysis (2015-2024). https://www.kaggle.com/datasets/atharvasoundankar/global-housing-market-analysis-2015-2024.

Mankiw, N. Gregory. Macroeconomics. 12th ed., Worth Publishers, 2016.